MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, FLIP-FLOPS, MONOLITHIC SILICON

This specification is approved for use by Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, bistable logic microcircuits. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. Device type shall be as shown in the following:

Device type	Circuit
01	Hex, D-type, positive-edge-triggered, flip-flops, with clear and single outputs.
02	Quad, D-type, positive-edge-triggered, flip-flops, with clear and complementary outputs.

1.2.2 Device class. Device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, MIL-M-38510, appendix C
E	D-2 (16-lead, 1/4" x 7/8" dual-in-line pack)
F	F-5 (16-lead, 1/4" x 3/8" flat pack)

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 to 7.0 Vdc
Input voltage range	-1.5 Vdc at -12 mA to 5.5 Vdc
Storage temperature range	-65° to 150°C
Maximum power dissipation per flip-flop, P_D 1/	73 mW (device type 01) 65 mW (device type 02)
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction to case	θ_{JC} = 0.09°C/mW for flat pack 0.08°C/mW for dual-in-line pack
Junction temperature	T_J = 175°C

FSC 5962

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1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

1.4 Recommended operating conditions.

Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.8 Vdc
Normalized fanout (each output) 2/	10 maximum
Ambient operating temperature range	-55° to 125°C
Input set up time	
Data input	25 ns
Clear inactive state	30 ns
Input hold time	5 ns

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal connections shall be as specified on figures 1 and 2, respectively.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 3.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 4.

3.2.4 Case outlines. Case outlines shall be as specified in 1.2.3 herein.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

2/ Device will fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V; I_{OH} = -800 µA	All	2.4	---	Volts
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V; I_{IN} = 16 mA	All	---	0.4	Volts
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IN} = -12 mA; T_A = 25°C	All	---	-1.5	Volts
Low-level input current	I_{IL1}	V_{CC} = 5.5 V; V_{IN} = 0.4 V 2/	All	-0.3	-1.6	mA
	I_{IL2}	V_{CC} = 5.5 V; V_{IN} = 0.4 V 3/	All	-0.4	-1.6	mA
	I_{IL3}	V_{CC} = 5.5 V; V_{IN} = 0.4 V 4/	All	-0.3	-0.8	mA
High-level input current	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.4 V	All	---	40	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	All	---	100	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V; V_{IN} = 0	All	-20	-57	mA
Supply current per device	I_{CC}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	01	---	65	mA
			02	---	45	mA
Maximum clock frequency	f_{MAX}	V_{CC} = 5 V; CL = 50 pF ±10%; RL = 390 Ω ±5%	All	25	---	MHz
Propagation delay to high logic level (clear to $\overline{Q}$)	t_{PLH1}		02	5	36	ns
Propagation delay to low logic level (clear to Q)	t_{PHL1}		All	5	50	ns
Propagation delay to high logic level (clock to Q)	t_{PLH2}		All	5	43	ns
Propagation delay to low logic level (clock to Q)	t_{PHL2}		All	5	43	ns
Propagation delay to high logic level (clock to $\overline{Q}$)	t_{PLH3}		02	5	43	ns
Propagation delay to low logic level (clock to $\overline{Q}$)	t_{PHL3}		02	5	43	ns

1/ See table III for complete terminal conditions.
2/ Clock input for device types 01 and 02.
3/ Clear input for device types 01 and 02.
4/ All D inputs for device types 01 and 02.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) Class A devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*, 2, 3, 7, 9	1*, 2, 3, 7, 9	1
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11	1, 2, 3, 7, 9	1, 2, 3, 9
Group C end point electrical parameters (method 5005)	N/A	1, 2, 3	1
Group D end point electrical parameters (method 5005)	1, 2, 3	1, 2, 3	1
Additional electrical subgroups for group C periodic inspections	None	10, 11	None

*PDA applies to subgroup 1 (see 4.3c.).

3.5 Rebonding. Rebonding shall be in accordance with MIL-M-38510.

3.6 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II. (Subgroups 7 and 8 testing requires only a summary of attributes data).

3.7 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. At the option of the manufacturer, the following marking may be omitted from the body of the microcircuit, but shall be retained on the initial container:

a. Country of origin.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 3 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

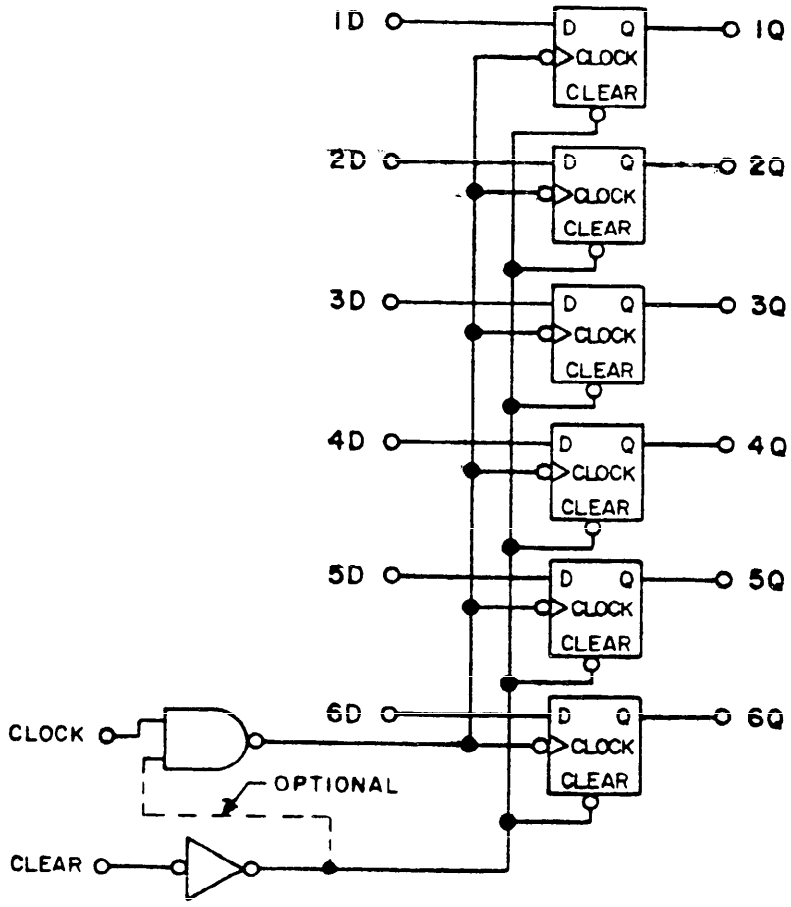
4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1, through 4.4.4).

4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

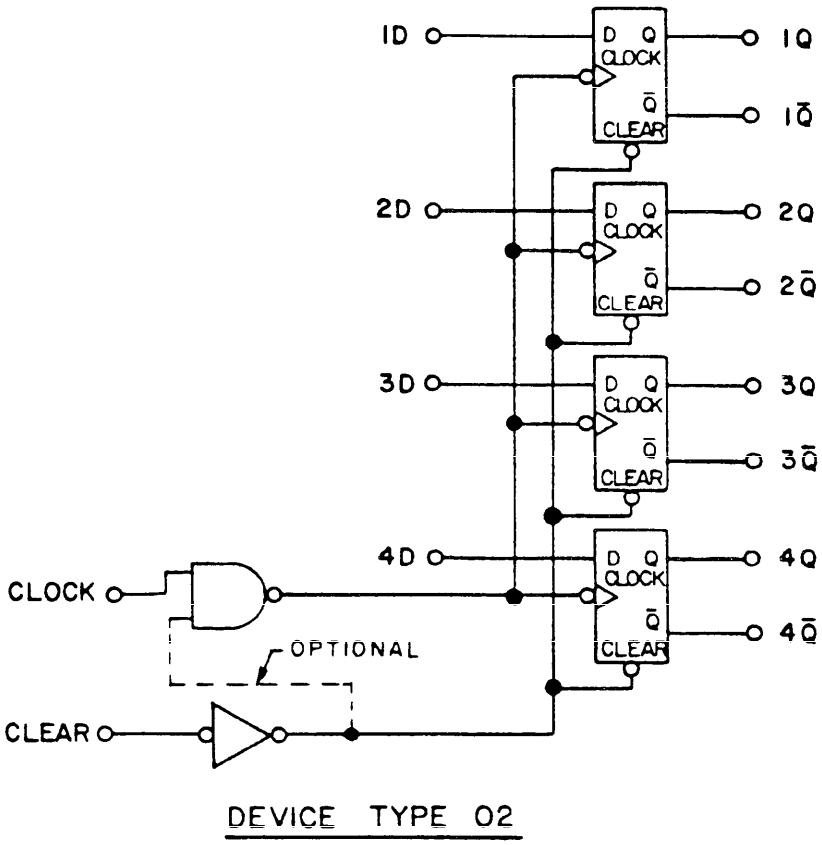
a. Burn-in test (method 1015 of MIL-STD-883).
   1. Test condition D or E, using the circuit shown on figure 5, or equivalent.
   2. T_A = 125°C minimum.
b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
c. Percent defective allowable (PDA) - The PDA for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).



DEVICE TYPE 01

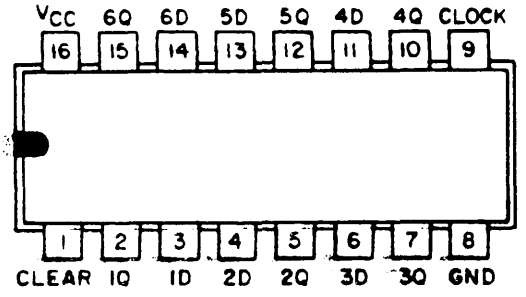
FIGURE 1. Logic diagrams.



DEVICE TYPE 02

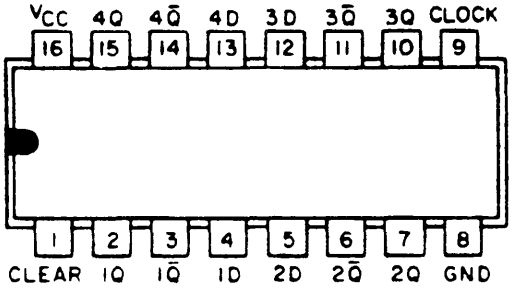
FIGURE 1. Logic diagrams - Continued.

Device type 01



CASES E AND F

Device type 02



CASES E AND F

FIGURE 2. Terminal connections.

TRUTH TABLE FOR EACH FLIP-FLOP				
CLEAR	CLOCK	D	Q	$\overline{Q}$ *
L	X**	X	L	H
H	↑	H	H	L
H	↑	L	L	H
H	L	X	Q_0	$\overline{Q}_0$

* DEVICE TYPE 02 ONLY
** INPUT MAY BE HIGH, LOW OR OPEN CIRCUIT

NOTE:
Clear is independent of clock. Information at the D input meeting the set-up time requirements is transferred to the Q output on the positive-going edge of the clock pulse. Clock triggering occurs at a particular voltage level and is not directly related to the transition time of the positive-going pulse. When the clock input is at either high or low level, the D input signal has no effect at the output.

FIGURE 3. Truth tables.

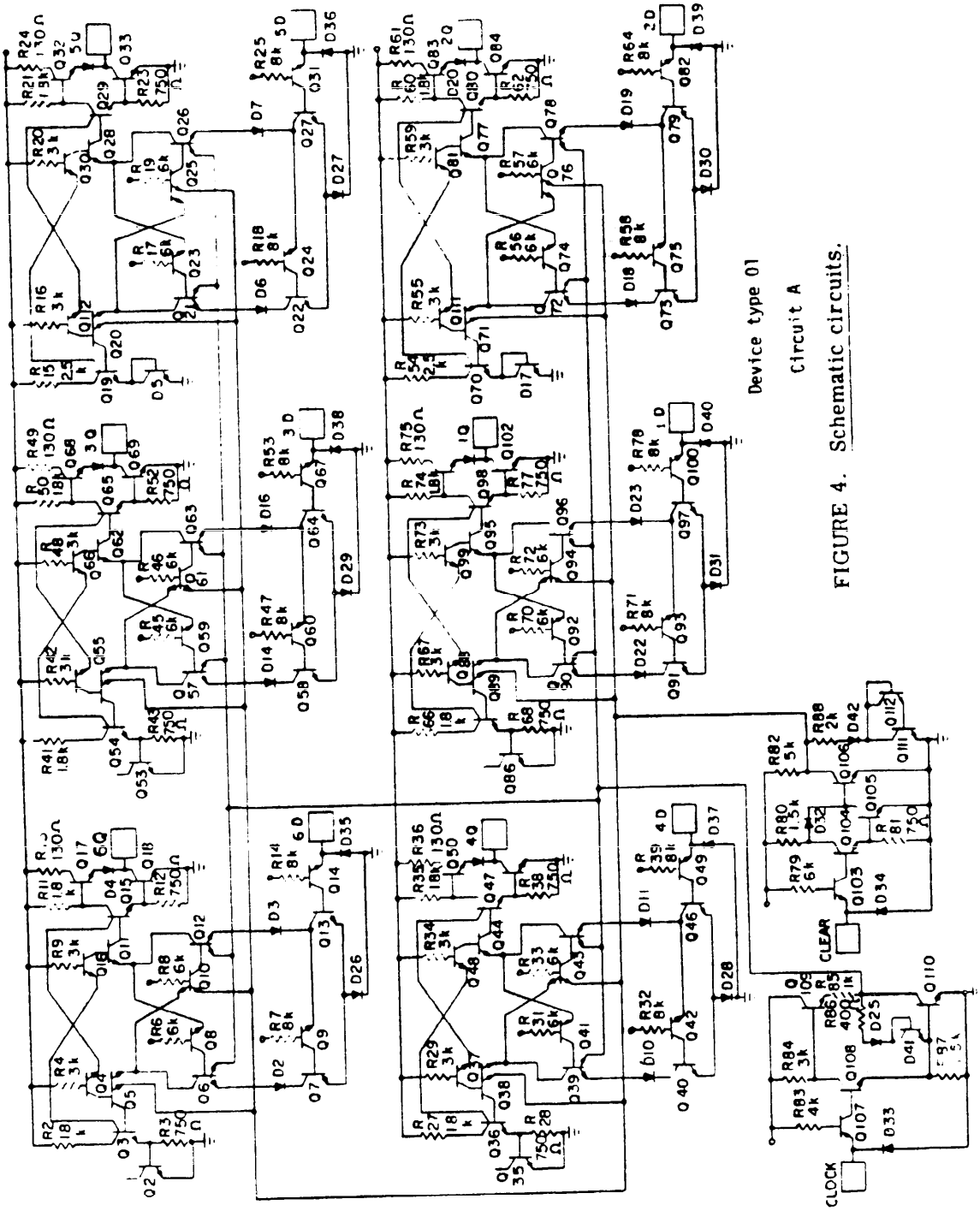
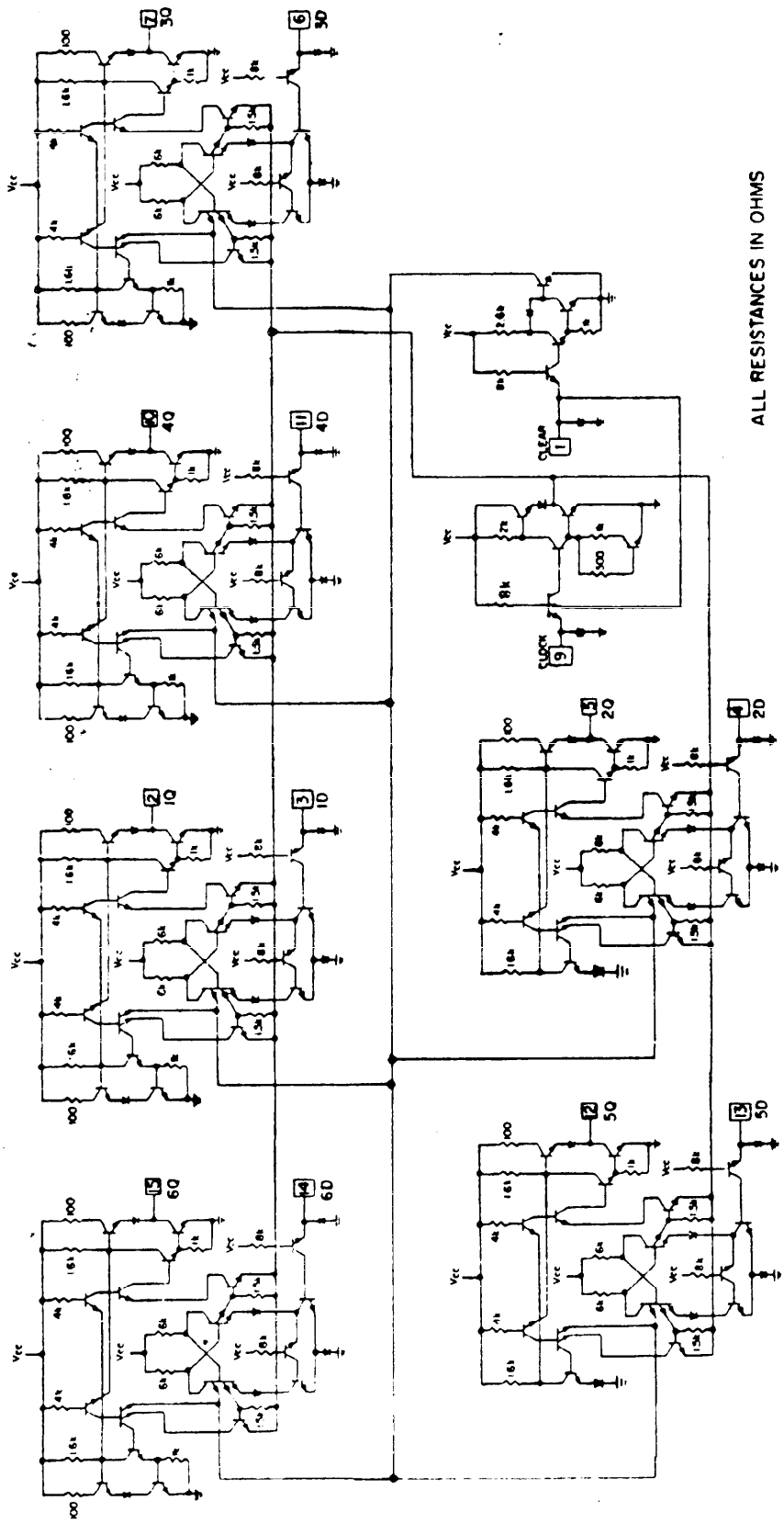
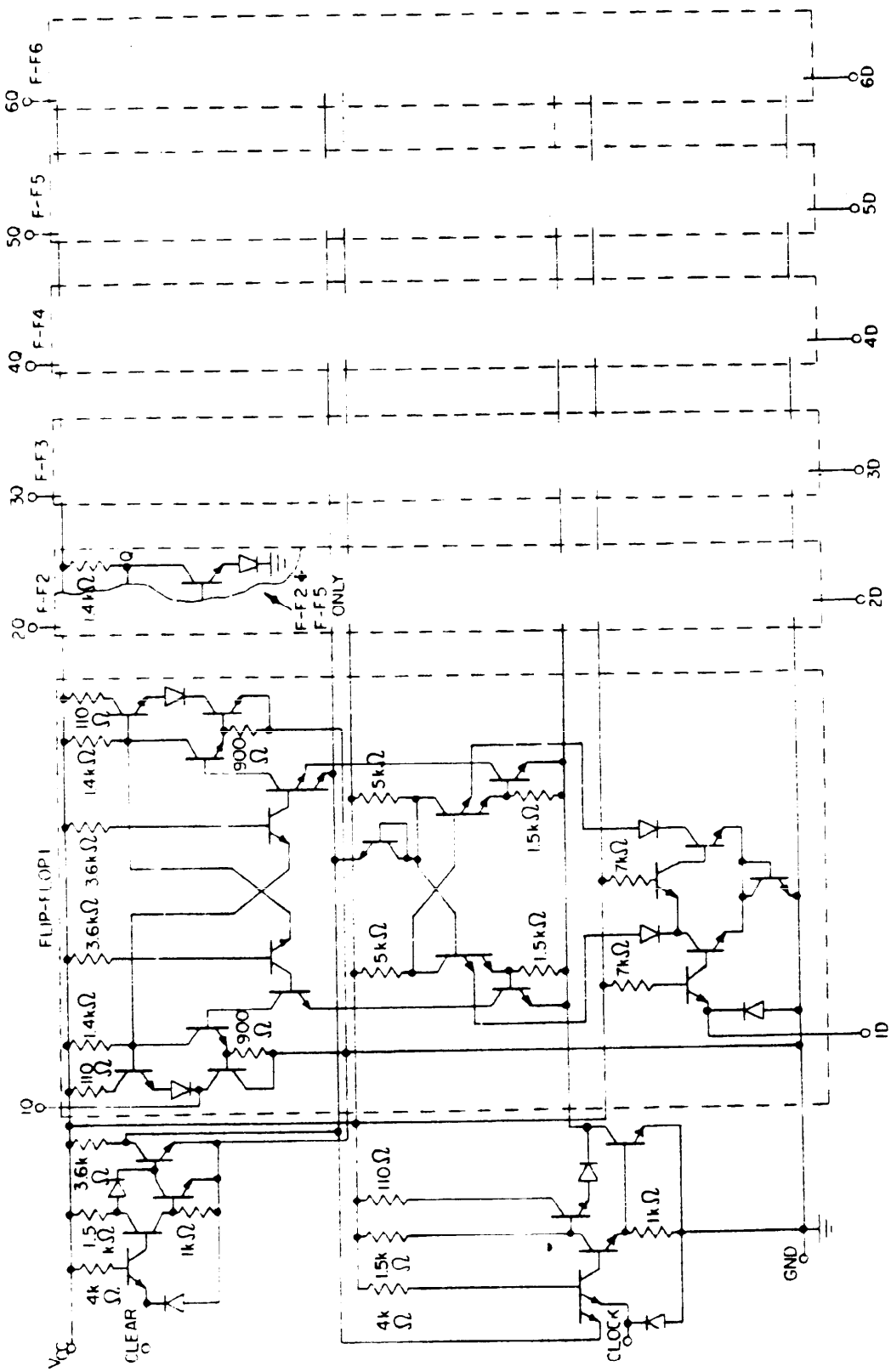


FIGURE 4. Schematic circuits.



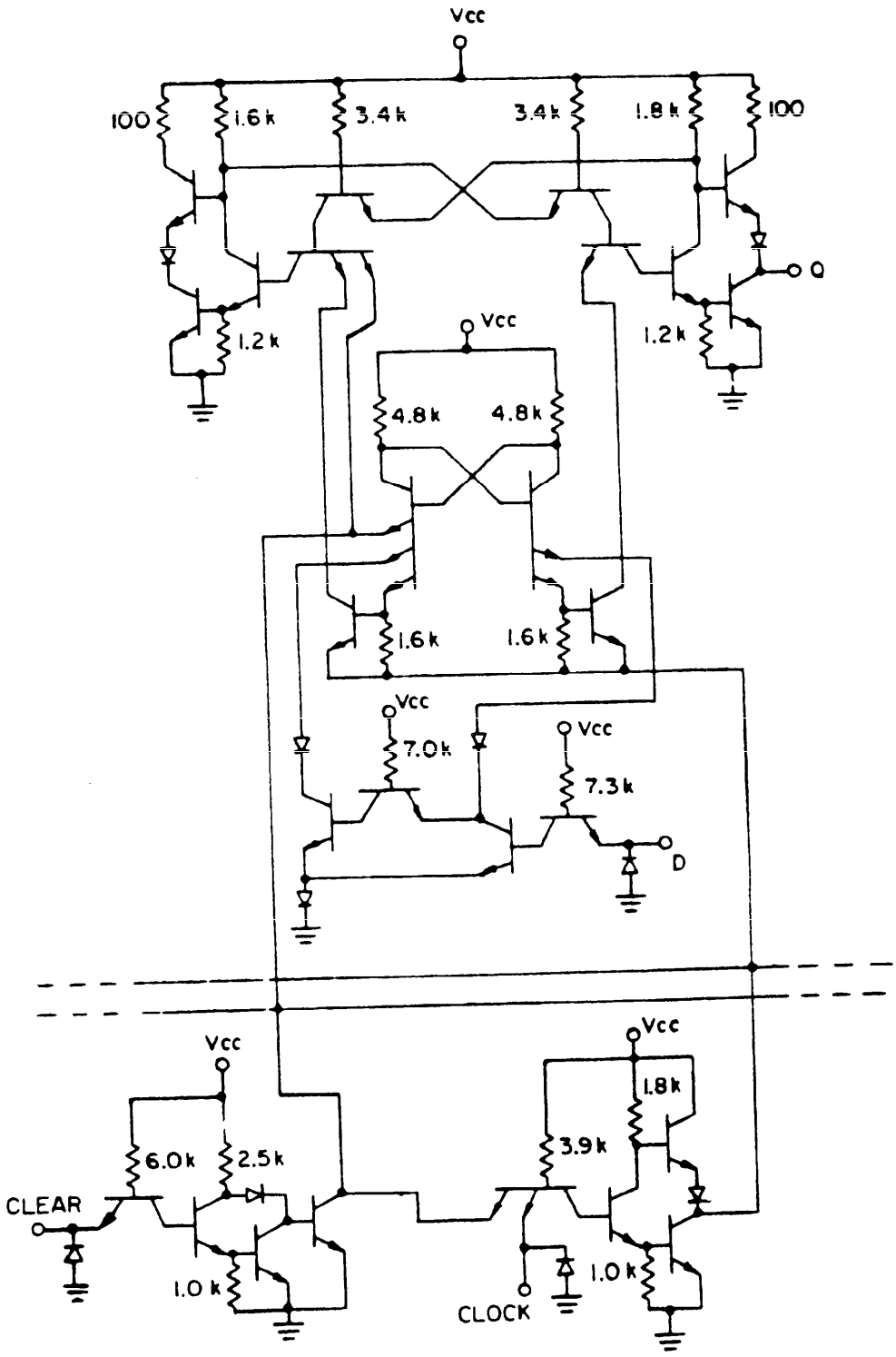
Device type 01 (Circuit B)

FIGURE 4. Schematic circuits - Continued.



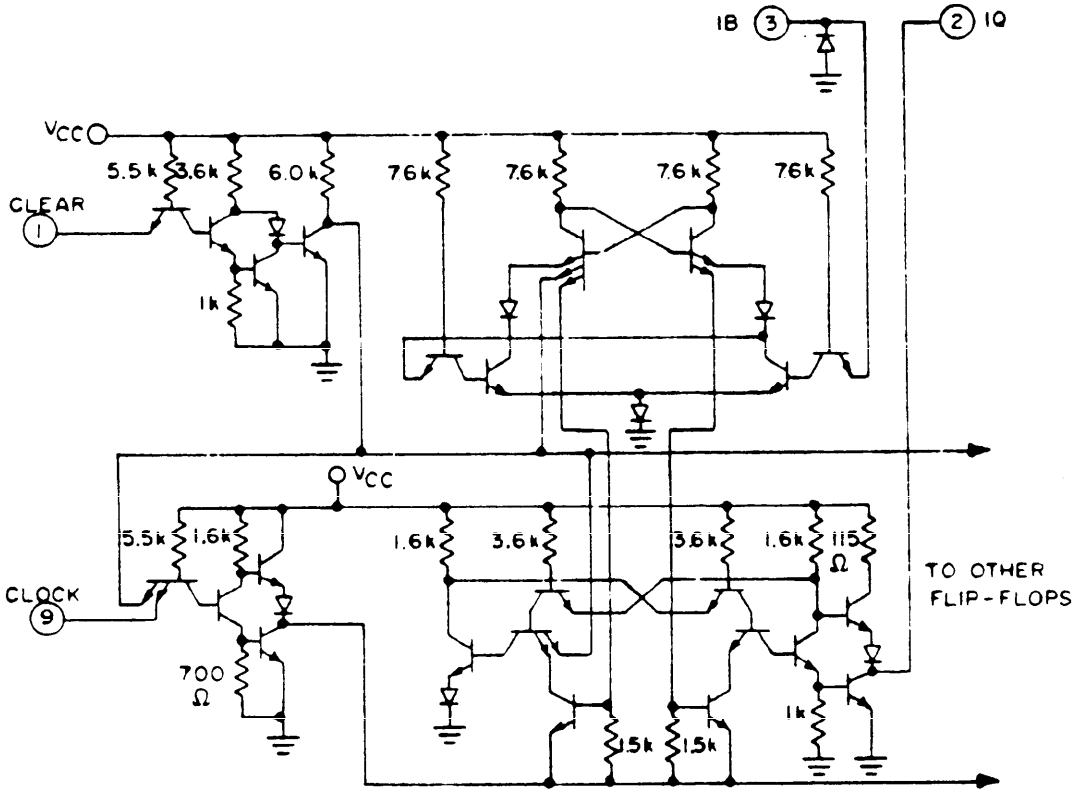
Device type 01 (Circuit C)

FIGURE 4. Schematic circuits - Continued.



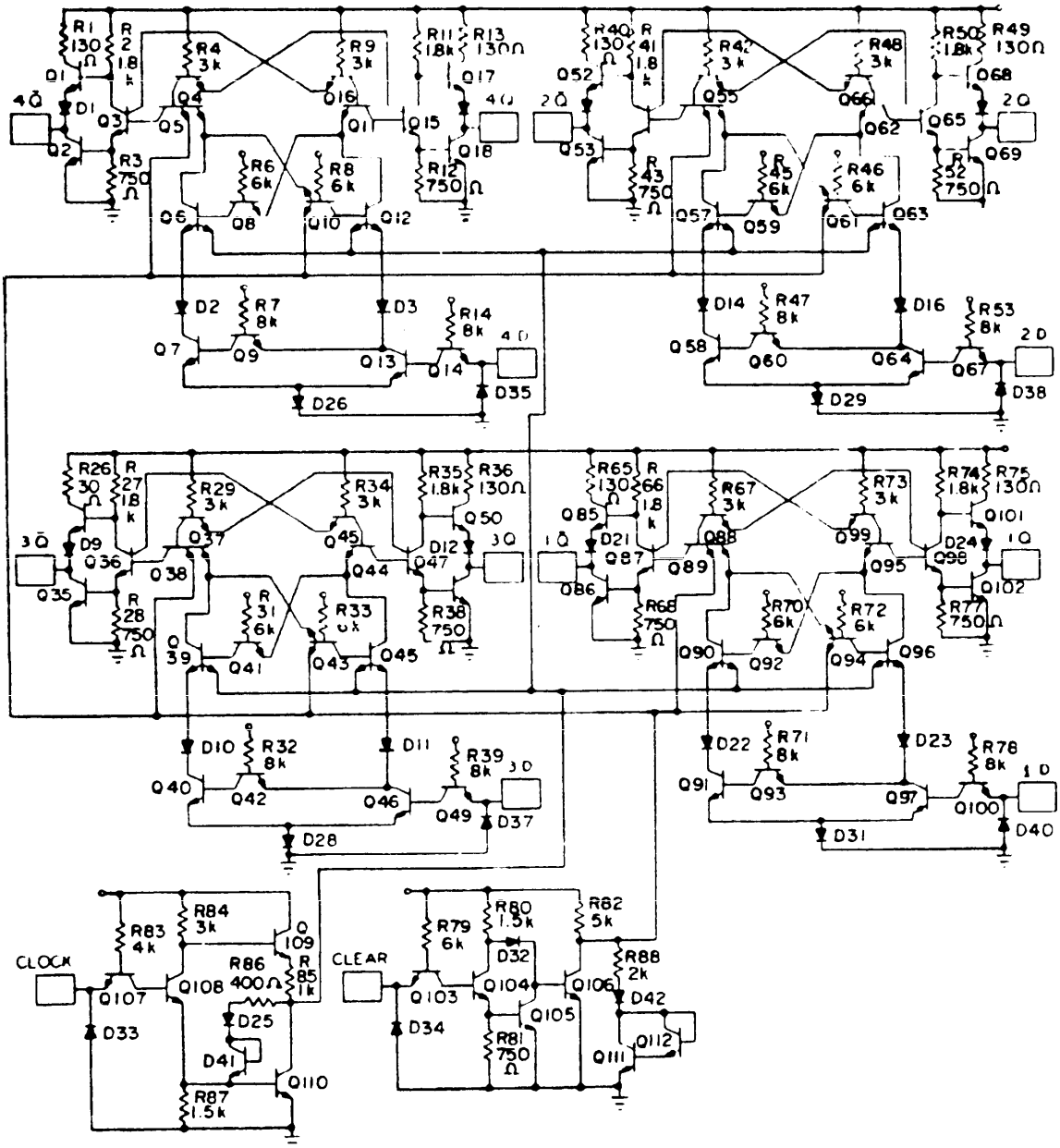
* Device type 01 (Circuit D)

FIGURE 4. Schematic circuits - Continued.



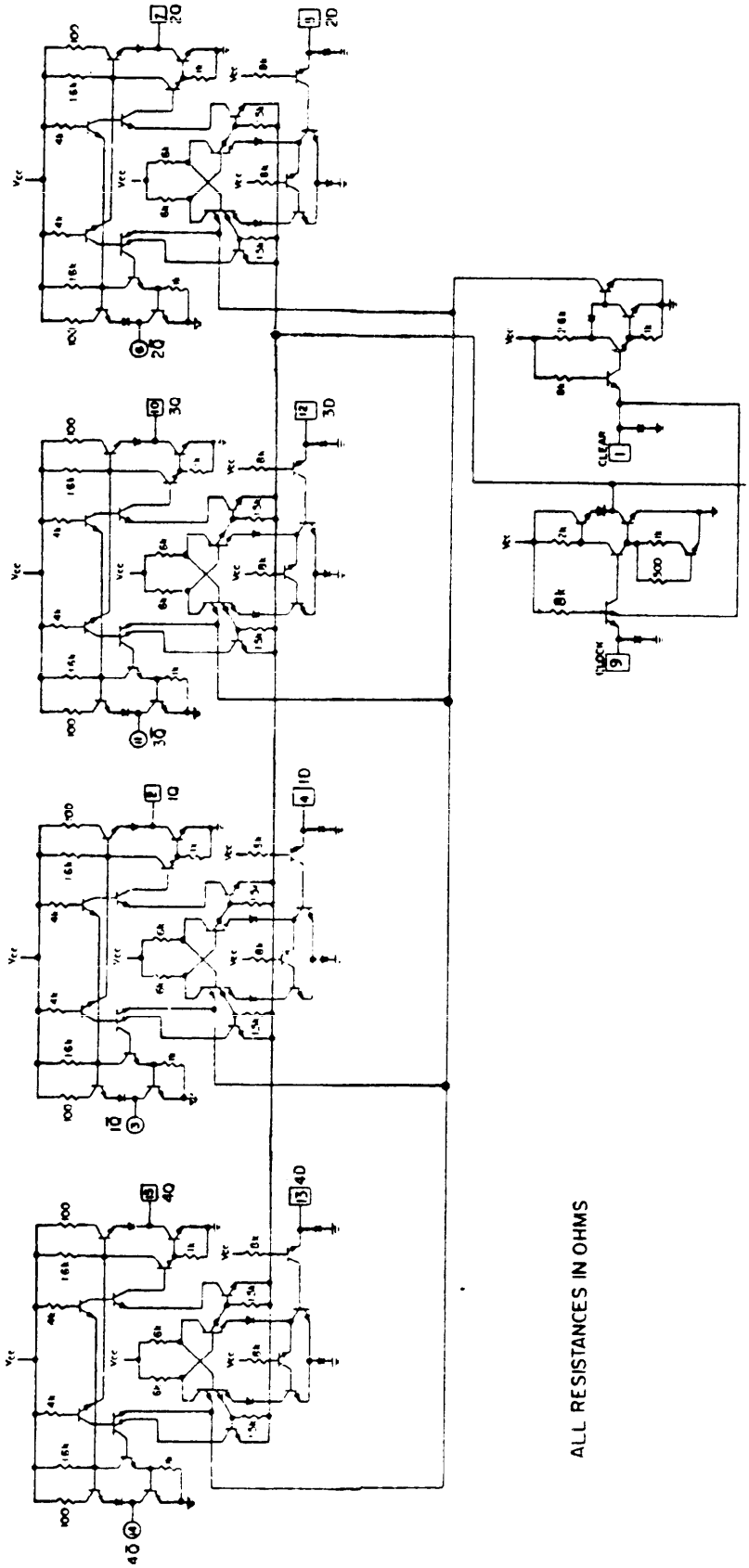
Device type 01 (Circuit E)

FIGURE 4. Schematic circuits - Continued.



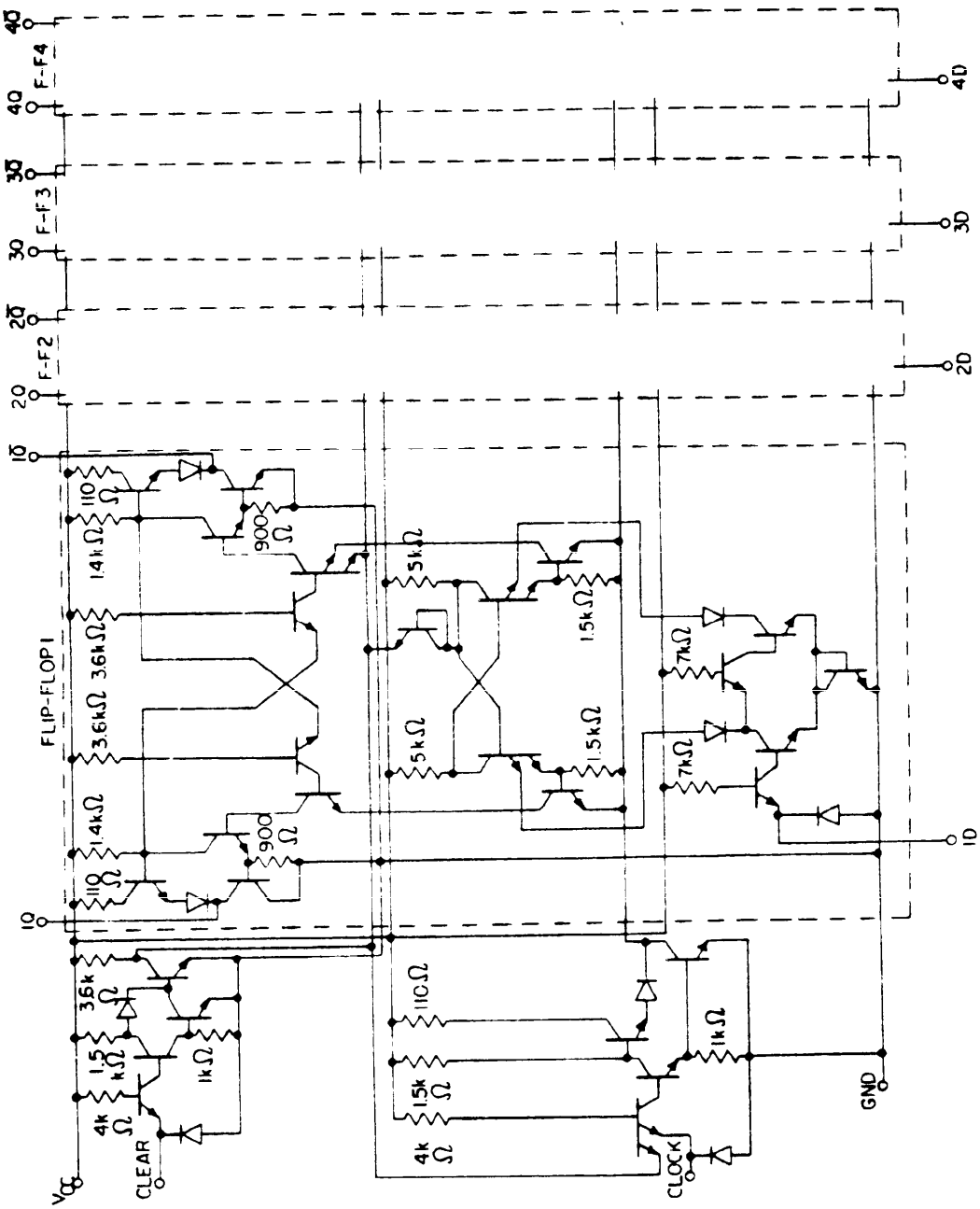
Device type 02
Circuit A

FIGURE 4. Schematic circuits - Continued.



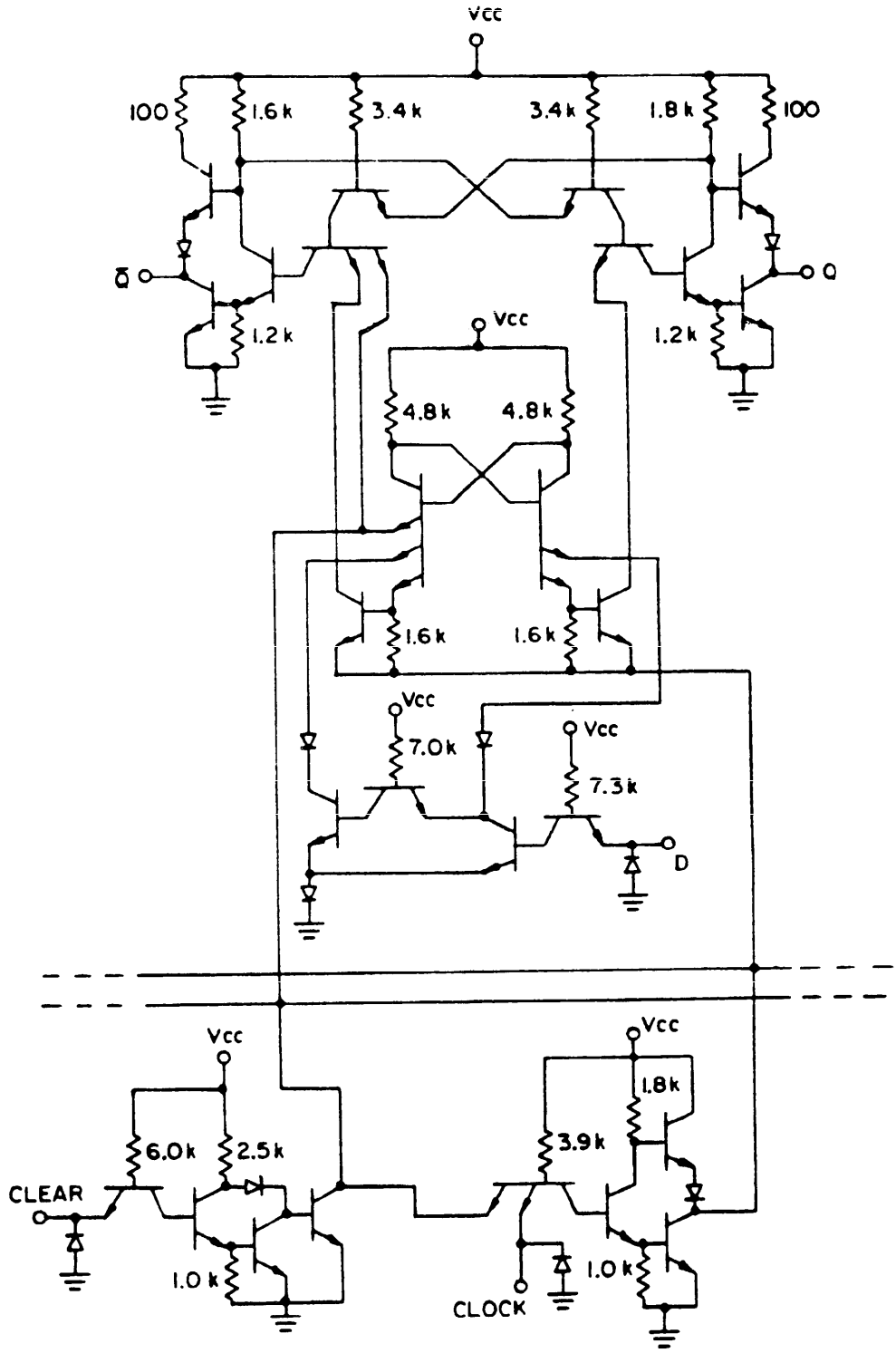
Device type 02 (Circuit B)

FIGURE 4. Schematic circuits - Continued.



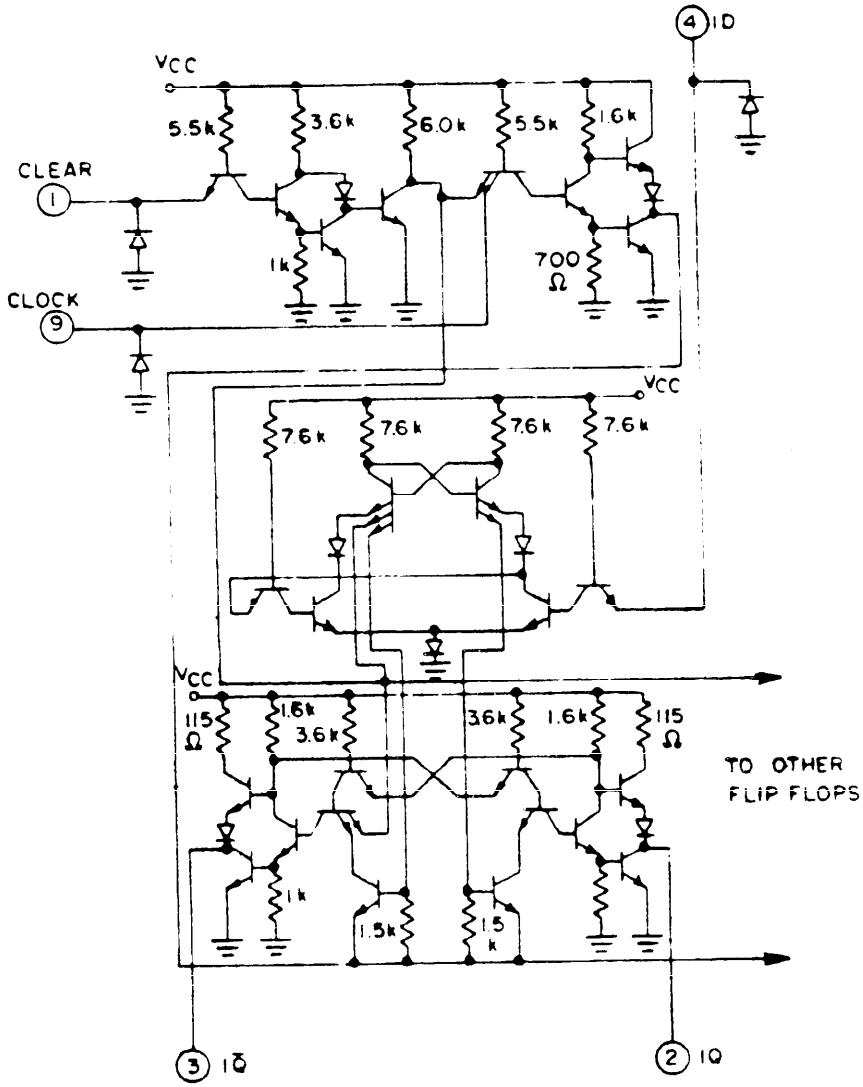
Device type 02 (Circuit C)

FIGURE 4. Schematic circuits



* Device type 02 (Circuit D)

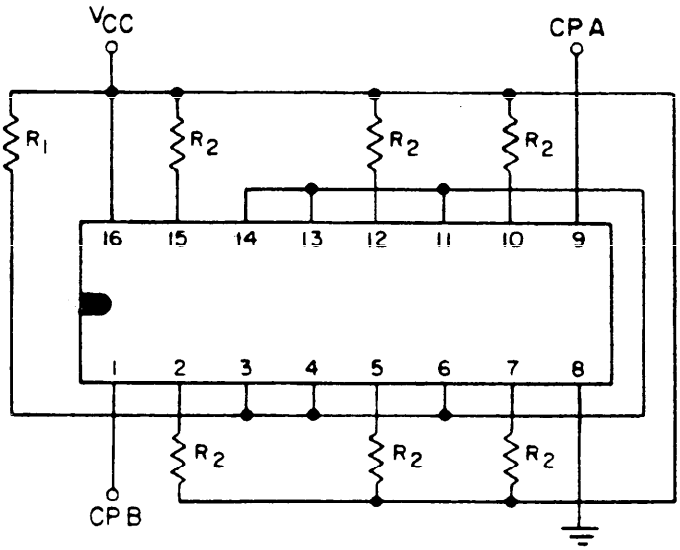
FIGURE 4. Schematic circuits - Continued.



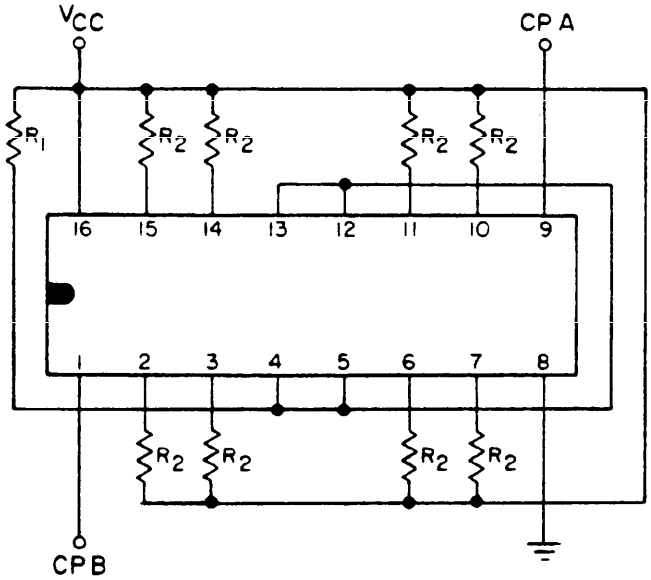
Device type 02 (Circuit E)

FIGURE 4. Schematic circuits - Continued.

Device type 01

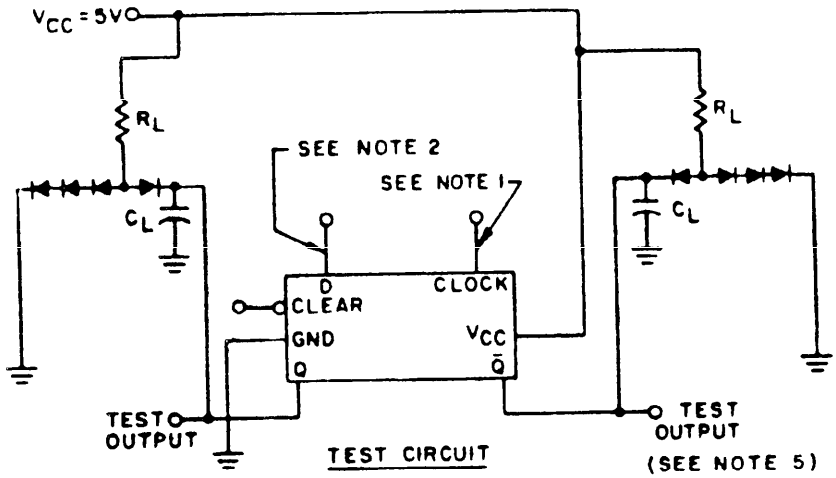


Device type 02

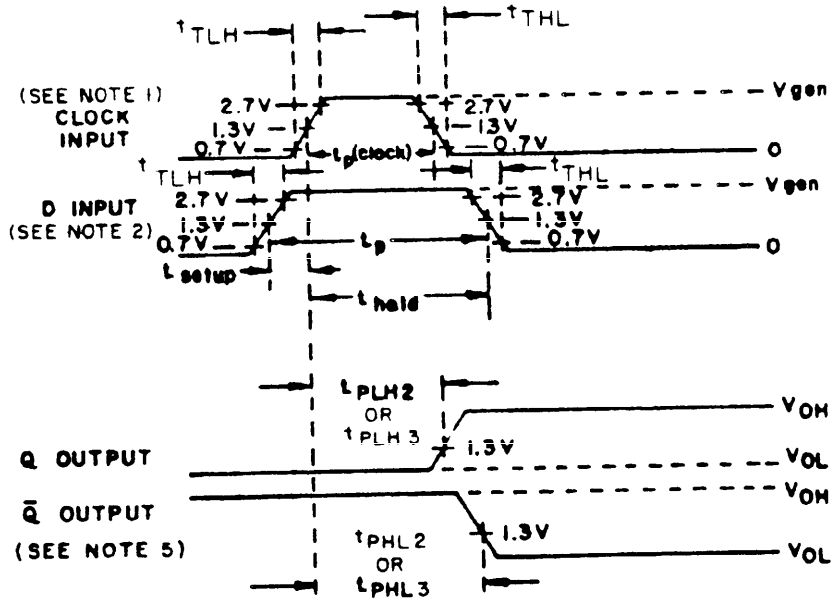


NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%: V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum. CPB = Same as CPA, synchronized with CPA, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R2 = 270Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 5. Burn-in and life test circuits.

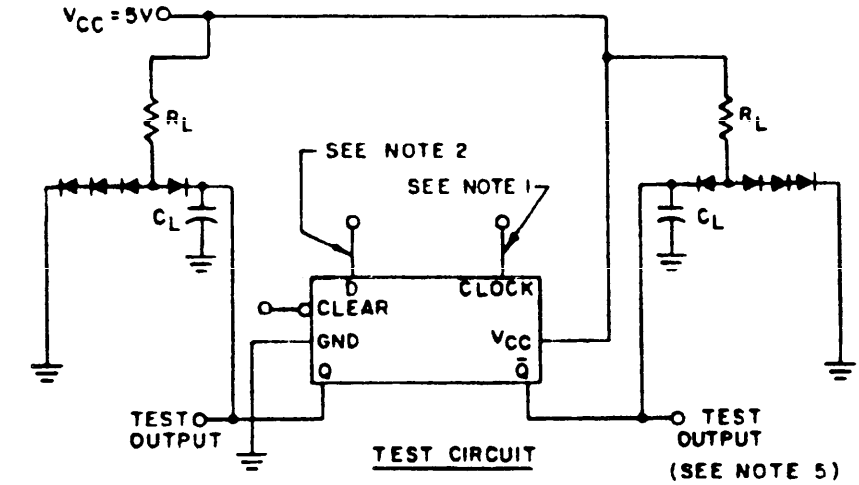


TEST CIRCUIT

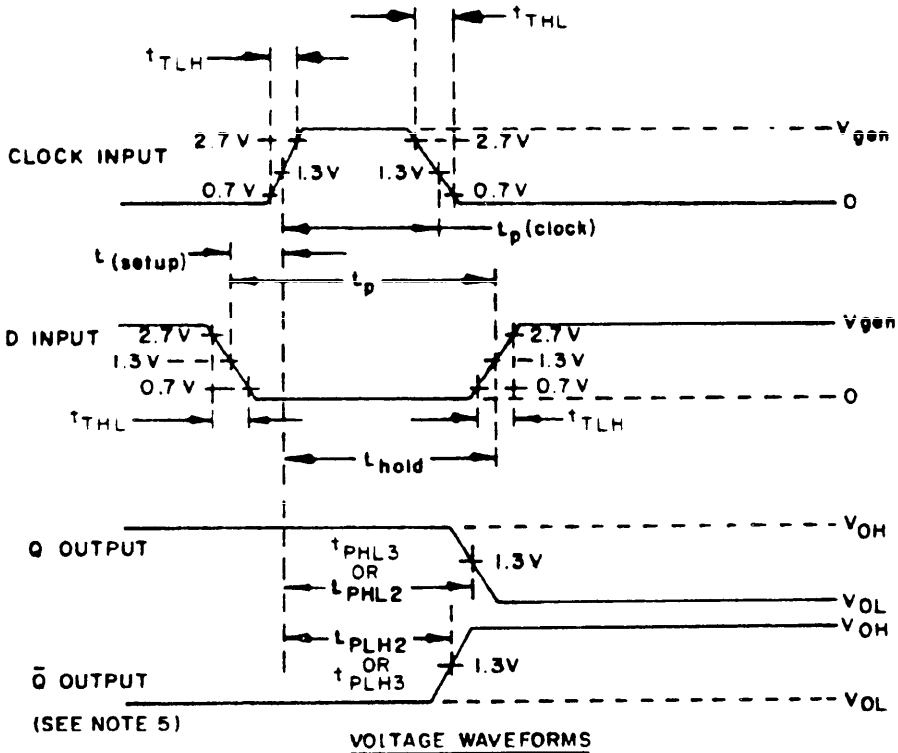


VOLTAGE WAVEFORMS

NOTES:
1. Clock input has the following characteristics: V_{gen} = 3 V minimum, t_p = 17 ns,



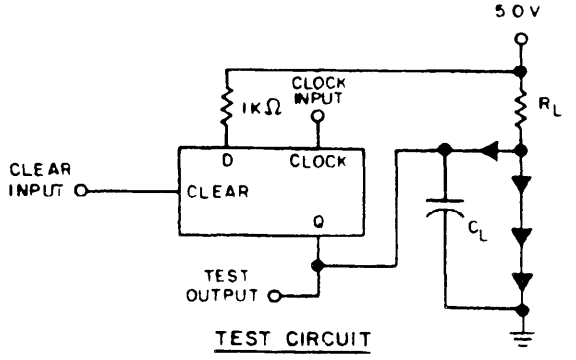
TEST CIRCUIT



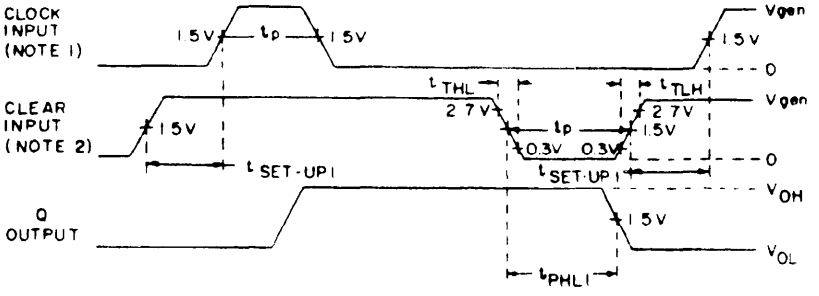
VOLTAGE WAVEFORMS

NOTES:
1. Clock input has the following characteristics: V_{gen} = 3 V minimum, t_p = 17 ns, $t_{THL} = t_{TLH} \leq 10$ ns, and PRR $\leq$ 1 MHz. When testing F_{MAX} PRR = 25 MHz.
2. D input pulse has the following characteristics: V_{gen} = 3 V minimum, $t_{TLH} = t_{THL} \leq 10$ ns, t_p = 30 ns, t_{SETUP} = 25 ns, t_{HOLD} = 5 ns and PRR < 0.5 MHz. When testing F_{MAX}, PRR = 12.5 MHz at 50% duty cycle.
3. R_L = 390Ω ±5%; C_L = 50 pF ±10% (including jig and probe capacitance).
4. All diodes are 1N3064, or equivalent.
5. $\bar{Q}$ output applies to device type 02 only.

FIGURE 7. Synchronous switching test circuit (low-level data) for device types 01 and 02.



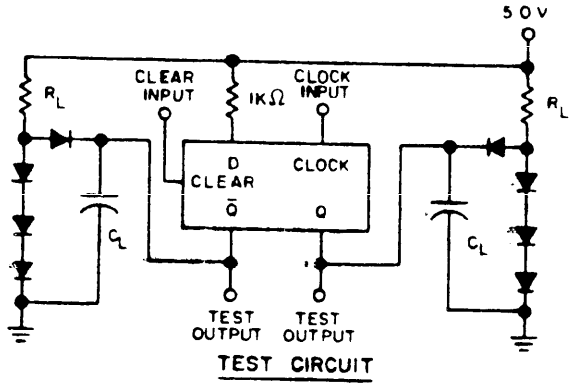
TEST CIRCUIT



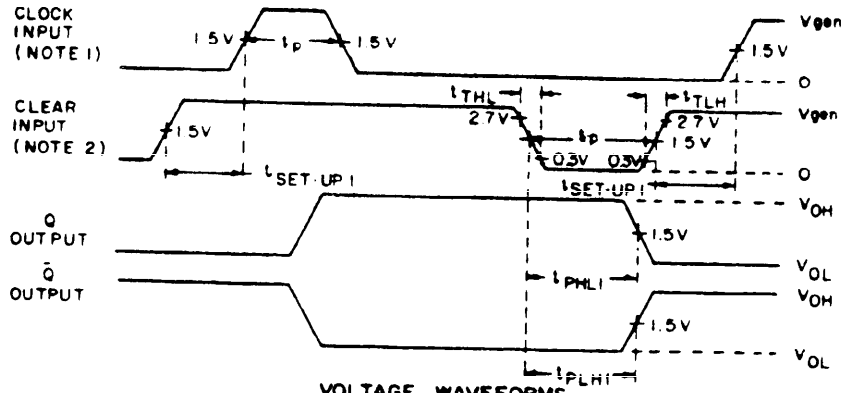
VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse is a preconditioning pulse and has the following characteristics: V_{gen} = 3 V minimum, $t_p \leq 100$ ns, t_{SETUP} = 25 ns and PRR $\leq$ 1 MHz.
2. Clear input pulse has the following characteristics: V_{gen} = 3 V minimum, $t_{THL} = t_{TLH} \leq 10$ ns, t_p = 30 ns, and PRR $\leq$ 1 MHz.
3. All diodes are 1N3064 or equivalent.
4. R_L = 390Ω ±5%; C_L = 50 pF ±10% (including jig and probe capacitance).

FIGURE 8. Clear switching test circuit and waveforms for device type 01.



TEST CIRCUIT



VOLTAGE WAVEFORMS

NOTES:
1. Clock input pulse is a preconditioning pulse and has the following characteristics: V_{gen} = 3 V minimum , $t_p \leq$ 100 ns, t_{SETUP} = 25 ns and PRR $\leq$ 1 MHz.
2. Clear input pulse has the following characteristics: V_{gen} = 3 V minimum , $t_{THL} = t_{TLH} \leq$ 10 ns, t_p = 30 ns, and PRR $\leq$ 1 MHz.
3. All diodes are 1N3064, or equivalent.
4. R_L = 390Ω ±5%; C_L = 50 pF ±10% (including jig and probe capacitance).

FIGURE 9. Clear switching test circuit and waveforms for device type 02.

Table III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case E & F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Max	Unit
				Clear	1Q	1D	2D	2Q	3D	3Q	GND	Clock	4Q	4D	5Q	5D	6D	6Q	V_{CC}				
1 T_A = 25°C	V_{IC}		1	-12mA							GND								4.5 V	Clear		-1.5	V
			2			-12 mA														1D			
			3				-12 mA													2D			
			4						-12 mA											3D			
			5									-12 mA								Clock			
			6											-12 mA						4D			
			7													-12 mA				5D			
			8														-12 mA			6D			
	V_{OH}	3007	9	2.0 V	-.8 mA	2.0 V						B								1Q	2.4		
			10				2.0 V	-.8 mA												2Q			
			11						2.0 V	-.8 mA										3Q			
			12										-.8 mA	2.0 V						4Q			
			13												-.8 mA	2.0 V				5Q			
			14														2.0 V	-.8 mA		6Q			
	V_{OL}	3008	15	0.8 V	16 mA															1Q		0.4	
			16					16 mA												2Q			
			17							16 mA										3Q			
			18										16 mA							4Q			
			19												16 mA					5Q			
			20															16mA		6Q			
	I_{IL1}	3009	21 CKT A,C,D									0.4 V							5.5 V	Clock	-0.7	-1.6	mA
			21 CKT B									0.4 V								Clock	-0.3	-0.8	
			21 CKT E									0.4 V								Clock	-0.4	-1.3	
	I_{IL2}		22 CKT A,D,E	0.4 V																Clear	-0.4	-1.3	
	I_{IL2}		22 CKT B,C	0.4 V																Clear	-0.7	-1.6	
	I_{IL3}		23			0.4 V														1D	-0.3	-0.8	
			24				0.4 V													2D			
			25						0.4 V											3D			
			26											0.4 V						4D			
			27													0.4 V				5D			
			28														0.4 V			6D			
	I_{IH1}	3010	29	2.4 V																Clear		40	μA
			30			2.4 V														1D			
			31				2.4 V													2D			
			32						2.4 V											3D			
			33									2.4 V								Clock			
			34											2.4 V						4D			
			35													2.4 V				5D			
			36														2.4 V			6D			
	I_{IH2}	3010	37	5.5 V																Clear		100	μA
			38			5.5 V														1D			
			39				5.5 V													2D			
			40						5 5 V											3D			
			41									5.5 V								Clock			
			42											5.5 V						4D			
			43													5.5 V				5D			
			44														5.5 V			6D			
	I_{OS}	3011	45	5.5 V	GND	5.5 V						B								1Q	-20	-57	mA
			46				5.5 V	GND												2Q			
			47						5.5 V	GND										3Q			
			48										GND	5.5 V						4Q			
			49												GND	5.5 V				5Q			
			50														5.5 V	GND		6Q			
	I_{CC}	3005	51	5.5 V		5.5 V	5.5 V		5.5 V			B		5.5 V		5.5 V	5.5 V			V_{CC}		65	

See notes at end of device type 01.

TABLE III. Group A inspection for device type 01 - Continued

Terminal conditions (pins not designated may be high ≥ 2.0 V or low ≤ 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case E & F Test No.	1 Clear	2 1Q	3 1D	4 2D	5 2Q	6 3D	7 3Q	8 GND	9 Clock	10 4Q	11 4D	12 5Q	13 5D	14 6D	15 6Q	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
2	Same tests, terminal conditions and limits as subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																						
7, 2, 4 T_A = 25°C	Truth table test	3014	52	B	L	A	A	L	A	H	GND	B	H	A	L	A	A	L	5.0 V	All outputs	Limits per 3		
			53	B								A											
			54	B								B											
			55	A								B											
			56									B											
			57		H			H		H		A	H		H			H					
			58		H	B	B	H	B	H		A	H	B	H	B	B	H					
			59		H	B	B	H	B	H		B	H	B	H	B	B	H					
			60		L	B	B	L	B	L		A	L	B	L	B	B	L					
			61		L	A	A	L	A	L		B	L	A	L	A	A	L					
			62		H			H		H		A	H		H			H					
			63		H			H		H		B	H		H			H					
			64	B	L			L		L		B	L		L			L					
8, 2, 4	Repeat subgroup 7 at T_A = 125°C and T_A = -55°C.																						
9 T_A = 25°C	F_{MAX} 5	(Fig 8)	65	5.0 V	OUT	IN	IN	OUT	IN	OUT	GND	IN	OUT	IN	OUT	IN	IN	OUT	5.0 V	1Q	25		MHz
			66																	2Q			
			67																	3Q			
			68																	4Q			
			69																	5Q			
			70																	6Q			
	t_{PHL1}	3003 (Fig 8)	71	IN	OUT	5.0 V	5.0 V	OUT	5.0 V	OUT			OUT	5.0 V	OUT	5.0 V	5.0 V	OUT		Clear-1Q	5	38	ns
			72																	Clear-2Q			
			73																	Clear-3Q			
			74																	Clear-4Q			
			75																	Clear-5Q			
			76																	Clear-6Q			
	$(t_{PLH2}$ $t_{PHL2})$	3003 (Fig 8&7)	77 & 78	5.0 V	OUT	IN	IN	OUT	IN	OUT			OUT	IN	OUT	IN	IN	OUT		Clock-1Q		33	
			79 & 80																	Clock-2Q			
			81 & 82																	Clock-3Q			
			83 & 84																	Clock-4Q			
			85 & 86																	Clock-5Q			
			87 & 88																	Clock-6Q			
10 T_A = 125°C	F_{MAX} 5	(Fig 8)	89	5.0 V	OUT	IN	IN	OUT	IN	OUT			OUT	IN	OUT	IN	IN	OUT		1Q	25		MHz
			90																	2Q			
			91																	3Q			
			92																	4Q			
			93																	5Q			
			94																	6Q			
	t_{PHL1}	3003 (Fig 8)	95	IN	OUT	5.0 V	5.0 V	OUT	5.0 V	OUT			OUT	5.0 V	OUT	5.0 V	5.0 V	OUT		Clear-1Q	5	50	ns
			96																	Clear-2Q			
			97																	Clear-3Q			
			98																	Clear-4Q			
			99																	Clear-5Q			
			100																	Clear-6Q			
	$(t_{PLH2}$ $t_{PHL2})$	3003 (Fig 8&7)	101 & 102	5.0 V	OUT	IN	IN	OUT	IN	OUT			OUT	IN	OUT	IN	IN	OUT		Clock-1Q		43	
			103 & 104																	Clock-2Q			
			105 & 106																	Clock-3Q			
			107 & 108																	Clock-4Q			
			109 & 110																	Clock-5Q			
			111 & 112																	Clock-6Q			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C.																						

NOTES:

1. B = Momentary GND, then V_{CC}.
2. The tests in subgroups 7 and 8 shall be performed in the sequence specified.
3. Output voltages shall be either:
   (a) H = 3.4 V minimum and L = 0.4 V maximum when using a high-speed checker double comparator, or
   (b) H > 1.5 V and L < 1.5 V when using a high-speed checker single comparator.
4. Only a summary of attributes data is required.
5. F_{MAX}, minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

TABLE III. Group A inspection for device type 02.

Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case E & F Test No.	1 Clear	2 1Q	3 1$\overline{Q}$	4 1D	5 2D	6 2$\overline{Q}$	7 2Q	8 GND	9 Clock	10 3Q	11 3$\overline{Q}$	12 3D	13 4D	14 4$\overline{Q}$	15 4Q	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	V_{IC}		1	-12 mA			-12 mA				GND								4.5 V	Clear		-1.5	V
			2					-12 mA												1D			
			3																	2D			
			4									-12 mA								Clock			
			5												-12 mA					3D			
			6													-12 mA				4D			
	V_{OH}	3007	7	2.0 V	-.8 mA		2.0 V					$\overline{B}$								1Q	2.4 V		
			8					2.0 V		-.8 mA					2.0 V					2Q			
			9										-.8 mA			2.0 V		-.8 mA			3Q			
			10																	4Q			
			11	0.8 V		-.8 mA														1$\overline{Q}$			
			12						-.8 mA											2$\overline{Q}$			
			13											-.8 mA						3$\overline{Q}$			
			14														-.8 mA			4$\overline{Q}$			
	V_{OL}		15	2.0 V		16 mA	2.0 V					$\overline{B}$								1$\overline{Q}$		0.4	
			16					2.0 V	16 mA											2$\overline{Q}$			
			17											16 mA	2.0 V					3$\overline{Q}$			
			18													2.0 V	16 mA			4$\overline{Q}$			
			19	0.8 V	16 mA															1Q			
			20							16 mA										2Q			
			21										16 mA							3Q			
			22															16 mA	5.5 V	4Q			
	I_{IL1}	3009	23 CKT A,C,D									0.4 V								Clock	-0.7	-1.6	mA
			23 CKT B									0.4 V								Clock	-0.3	-0.8	
			23 CKT E									0.4 V								Clock	-0.4	-1.3	
	I_{IL2}		24 CKT A,D,E	0.4 V																Clear	-0.4	-1.3	
	I_{IL2}		24 CKT B,C	0.4 V																Clear	-0.7	-1.6	
	I_{IH3}		25				0.4 V													1D	-0.3	-0.8	
			26					0.4 V												2D			
			27												0.4 V					3D			
			28													0.4 V				4D			
	I_{IH1}	3010	29	2.4 V																Clear		40	μA
			30				2.4 V													1D			
			31					2.4 V												2D			
			32									2.4 V								Clock			
			33												2.4 V					3D			
			34													2.4 V				4D			
	I_{IH2}	3010	35	5.5 V																Clear		100	μA
			36				5.5 V													1D			
			37					5.5 V												2D			
			38									5.5 V								Clock			
			39												5.5 V					3D			
			40													5.5 V				4D			
	I_{OS}	3011	41	5.5 V	GND		5.5 V					$\overline{B}$								1$\overline{Q}$	-20	-57	mA
			42					5.5 V		GND					5.5 V					2$\overline{Q}$			
			43										GND			5.5 V		GND		3$\overline{Q}$			
			44																	4$\overline{Q}$			
			45	GND		GND														1Q			
			46	GND					GND											2Q			
			47											GND						3Q			
			48														GND			4Q			
	I_{CC}	3005	49	5.5 V			5.5 V	5.5 V				$\overline{B}$			5.5 V	5.5 V				V_{CC}		45	
2	Same tests, terminal conditions and limits as subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																						

See notes at end of device type 02.

TABLE III. Group A inspection for device type 02. - Continued

Terminal conditions (pins not designated may be high = 2.0 V or low = 0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Case E & F Test No.	1 Clear	2 1Q	3 1$\overline{Q}$	4 1D	5 2D	6 2$\overline{Q}$	7 2Q	8 GND	9 Clock	10 3Q	11 3$\overline{Q}$	12 3D	13 4D	14 4$\overline{Q}$	15 4Q	16 V_{CC}	Meas. terminal	Test Limits Min	Max	Unit
7, 2, 4 T_A = 25°C	Truth table test	3014	50	B	I	H	A	A	H	I	GND	B	L	H	A	A	H	L	5.0 V	All outputs	Limits per 3		
			51	B	I	H			H			A	I	H			H	L					
			52	B	I	H			H			B	L	H			H	L					
			53	A	L	H			H			B	L	H			H	L					
			54		H	I			I	H		A	H	I			L	H					
			55		H	I			I	H		B	H	I			I	H					
			56		H	I			I	H		B	H	I	B	B	I	H					
			57		I	H	B	B	H	I		A	L	H	B	B	H	L					
			58		I	H	B	B	H	L		B	I	H	B	B	H	L					
			59		I	H	B	B	H	L		B	I	H	A	A	H	L					
			60		H	L	A	A	I	H		A	H	I	A	A	I	H					
			61		H	I	A	A	I	H		B	H	L	A	A	I	H					
			62	B	I	H	A	A	H	I		B	I	H	A	A	H	I					
8, 9 T_A = 25°C	Repeat subgroup 7 at T_A = 125°C and T_A = -55°C.																						
	f_{MAX} 5	(Fig 6)	63	5.0 V	OUT		IN	IN		OUT	GND	IN	OUT		IN	IN		OUT	5.0 V	1Q 2Q 3Q 4Q	25		MHz
			64																				
			65																				
			66																				
	t_{PLH1}	3003 (Fig 8)	67	IN		OUT	5.0 V	5.0 V	OUT			IN		OUT	5.0 V	5.0 V	OUT			Clear-1Q Clear-2Q Clear-3Q Clear-4Q	5	28	ns
			68																				
			69																				
			70																				
	t_{PHL1}		71		OUT		5.0 V	5.0 V		OUT			OUT		5.0 V	5.0 V		OUT		Clear-1Q Clear-2Q Clear-3Q Clear-4Q		38	
			72																				
			73																				
			74																				
	(t_{PLH2} t_{PHL2})	3003 Fig. 6&7	75 & 76		OUT		IN	IN		OUT			OUT		IN	IN		OUT		Clock-1Q Clock-2Q Clock-3Q Clock-4Q		33	
			77 & 78																				
			79 & 80																				
			81 & 82																				
	(t_{PLH3} t_{PHL3})	3003 Fig. 6&7	83 & 84			OUT	IN	IN	OUT					OUT	IN	IN	OUT			Clock-1Q Clock-2Q Clock-3Q Clock-4Q			
			85 & 86																				
			87 & 88																				
			89 & 90																				
10 T_A = 125°C	f_{MAX} 5	(Fig 6)	91	5.0 V	OUT		IN	IN		OUT	GND		OUT		IN	IN		OUT	5.0 V	1Q 2Q 3Q 4Q	25		MHz
			92																				
			93																				
			94																				
	t_{PLH1}	3003 (Fig 8)	95	IN		OUT	5.0 V	5.0 V	OUT					OUT	5.0 V	5.0 V	OUT			Clear-1Q Clear-2Q Clear-3Q Clear-4Q	5	38	ns
			96																				
			97																				
			98																				
	t_{PHL1}		99		OUT		5.0 V	5.0 V		OUT			OUT		5.0 V	5.0 V		OUT		Clear-1Q Clear-2Q Clear-3Q Clear-4Q		50	
			100																				
			101																				
			102																				
	(t_{PLH2} t_{PHL2})	3003 Fig. 6&7	103 & 104		OUT		IN	IN		OUT			OUT		IN	IN		OUT		Clock-1Q Clock-2Q Clock-3Q Clock-4Q		43	
			105 & 106																				
			107 & 108																				
			109 & 110																				
	(t_{PLH3} t_{PHL3})	3003 Fig. 6&7	111 & 112			OUT	IN	IN	OUT					OUT	IN	IN	OUT			Clock-1Q Clock-2Q Clock-3Q Clock-4Q			
			113 & 114																				
			115 & 116																				
			117 & 118																				
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C.																						

NOTES:

1. B = Momentary GND, then V_{CC}.
2. The tests in subgroups 7 and 8 shall be performed in the sequence specified.
3. Output voltages shall be either
   (a) H = 2.4 V minimum and L = 0.4 V maximum when using a high-speed checker double comparator, or
   (b) H = 1.5 V and L = 1.5 V when using a high-speed checker single comparator.
4. Only a summary of attributes data is required.
5. f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II.
b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

4.4.3 Group C inspections. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.
b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.
c. Operating life test (method 1005 of MIL-STD-883) conditions, or equivalent:
   1. Test condition D or E, using the circuit shown on figure 5, or equivalent.
   2. T_A = 125°C, minimum.
   3. Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate tables and as follows:

4.5.1 Life-test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias. Alternately, the bias may be removed during cooling if the case temperature is reduced to room temperature within 30 minutes after removal of the test condition.

4.6 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-M-38510, except that the rough handling test shall not apply.

5. PACKAGING

5.1 Packaging requirements. The requirements for the packaging of microcircuits shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logic support.

6.3 Ordering data. The contract or order should specify the following:

a. Complete part number (see 1.2).
b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
c. Requirement for certificate of compliance, if applicable.
d. Requirements for notification of change of product or process to procuring activity in addition to notification to qualifying activity, if applicable.
e. Requirements for preservation and packing.
f. Requirements for failure analysis (including required test condition of method 5003), corrective action, and reporting of results, if applicable.
g. Requirements for product assurance options.
h. Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-STD-1313, MIL-STD-1331, and as follows:

GND - - - - - - - Electrical ground (common terminal)
V_{IN} - - - - - - - Voltage level at an input terminal

6.4.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional current and positive when flowing into the referenced terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed Generic-Industry type. Generic-Industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of Generic-Industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military Device type	Generic-Industry type
01	54174
02	54175

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 85, 99

User activity:
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-F295)

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

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